

Homoscedasticity and Its Testing: From Conditional Expectation to the Breusch–Pagan Test

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Part 9 / 12
Sequential Sampling and the Bayesian Connection

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1 Sequential Sampling of Random Quantities

1.1 Motivation

We routinely manipulate measure-dependent functions: we sample them, integrate them, take moments, differentiate them, form derived quantities. The density f_P of a measure P is one such object. A natural question: can we iterate this process — plug a random draw back into the density, treat the result as a new random variable, and repeat? This section explores three such constructions and their limiting behaviour.

1.2 Setup

Fix a probability measure P on $\mathbb{R}$ with Lebesgue density $f = f_P \geq 0$, $\int f = 1$. Draw $X_1 \sim P$. Since $f(x) \geq 0$ and $\int f = 1$, the function f maps $\mathbb{R}$ into $[0, \sup f]$, so all iterates stay bounded above by $\sup f < \infty$ (assuming the density is bounded, as for the normal family).

1.3 Three sequences

Sequence P_n (chained push-forward). Define $X_{n+1} = f_{P_n}(X_n)$, where P_n is the distribution of X_n and f_{P_n} is its density. Each step is a *push-forward*: $P_{n+1} = (f_{P_n})_* P_n$. The density changes at every step together with the sampling distribution.

Sequence Q_n (orbit of a point under f). Fix the original density $f = f_{P_1}$ throughout. Set

$$X_{n+1} = f(X_n), \quad X_1 \sim P.$$

This is deterministic given X_1 : the sequence $X_1, f(X_1), f(f(X_1)), \dots$ is the *orbit* of the initial point X_1 under the iterated application of f . The randomness enters only through the initial draw; thereafter Q_n is the push-forward of P under $f^{\circ n}$ (the n -fold composition of f with itself).

Sequence R_n (fixed sample point, varying density). Keep X_1 fixed and let the density vary. Concretely, if P_n has density f_n , form

$$R_n = f_n(X_1).$$

Here X_1 is a fixed (random) argument and f_n evolves. This is closer to evaluating a likelihood at a fixed data point under a changing model.

1.4 Limiting behaviour of sequence Q_n

The sequence Q_n is the most concrete: it reduces to the dynamics of the map $f : \mathbb{R} \rightarrow [0, \sup f]$.

Contraction. After one step, every point is mapped into $[0, \sup f]$. If f is a contraction on this interval — i.e. $|f'(x)| < 1$ for all x in the range — then by the Banach fixed-point theorem the orbit converges to the unique fixed point x^* satisfying

$$f(x^*) = x^*.$$

Fixed point for the normal density. For $f(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$ the equation $f(x^*) = x^*$ has a solution near $x^* \approx 0.3989$. (At $x = 0$: $f(0) = 1/\sqrt{2\pi} \approx 0.399 > 0$; at $x = 1$: $f(1) \approx 0.242 < 1$; by the intermediate value theorem a crossing exists.) Moreover $f'(x) = -xf(x)$, so $|f'(x^*)| = x^* \cdot f(x^*) = (x^*)^2 \approx 0.159 < 1$: the fixed point is attracting and the orbit converges to it for almost every starting point X_1 .

The limiting distribution of Q_n therefore degenerates to a point mass at x^* : starting from any $X_1 \sim P$, the sequence $f(X_1), f(f(X_1)), \dots$ converges almost surely to x^* .

Remark 1 (Degenerate limit). The limit measure is δ_{x^*} , a Dirac mass. It has no Lebesgue density. So even if every Q_n has a density, the limit can lose absolute continuity.

1.5 Limiting behaviour of sequence P_n

The chained push-forward is more complex because the density changes at each step. Even in the normal case, P_2 is the distribution of $f(X_1)$ where $X_1 \sim \mathcal{N}(0, 1)$: its density is obtained by the change-of-variables formula applied to the non-monotone map f . Since f is even (symmetric about 0) and has a unique maximum at 0, the map is two-to-one on $\mathbb{R} \setminus \{0\}$ and the density of P_2 has support in $(0, 1/\sqrt{2\pi}]$.

For the general sequence P_n , the densities f_{P_n} contract their support and, under mild conditions, the sequence converges to a degenerate measure — the rate and limit depending on P .

1.6 Dependence on the original distribution P

All three sequences depend only on P . Different choices of P produce qualitatively different behaviour:

- **Light-tailed unimodal P** (e.g. normal): the density is bounded, the fixed point x^* exists and is attracting, the Q_n sequence converges to δ_{x^*} .
- **Heavy-tailed P** (e.g. Cauchy): the density is still bounded ($\sup f = 1/\pi$), but the orbit dynamics depend on the fixed-point equation for the Cauchy density.
- **Multimodal P** : the density map f may have multiple fixed points or even cycles, leading to more complex orbit structure (period-doubling, chaos).
- **Unbounded density** (e.g. Beta(1/2, 1/2), which diverges at 0 and 1): $\sup f = \infty$ and the contraction argument breaks down; the sequence may diverge.

1.7 Connection to Bayesian iteration and Markov chains

The construction in sequence P_n is reminiscent of Bayesian updating: starting from a prior, each observation shifts the measure. But here the “data” is the density itself evaluated at the current draw — a self-referential structure.

More formally, Q_n defines a (deterministic given X_1) dynamical system on $\mathbb{R}$. The randomness in X_1 induces a distribution over orbits. If f has a unique attracting fixed point x^* , the empirical measure of the orbit concentrates on δ_{x^*} regardless of the starting distribution P (as long as P assigns positive mass to the basin of attraction of x^* , which for the normal density is all of $\mathbb{R}$).

Remark 2 (Open questions). The limiting behaviour of P_n (chained push-forward) for general P is not fully characterised here. The key quantities are: (a) whether f_{P_n} has a fixed point; (b) the spectral radius of the linearisation Df_{P_n} at that fixed point; (c) whether the push-forward operation preserves absolute continuity. These depend on the smoothness and tail behaviour of P .